

ON WEAKER TOPOLOGICAL STRUCTURES AND σ -TOPOLOGICAL SPACES

E. CHERNYSH

ABSTRACT. In this article we weaken the notion of a topology and study resulting spaces. We shall contrast topological spaces to these new spaces and see elementary properties which do not carry over. Finally, we study such a space resulting from the σ -algebra of Lebesgue measurable sets in $\mathbb{R}^d$.

1. σ -SPACES

In this section we shall introduce the simple, yet considerable, notion of a “generalized” topology and their induced topological spaces. The main contrast with usual topological spaces lies in the union requirement of the topology $\mathcal{T}$: we shall require only closure under countable unions.

We shall first introduce the concept and provide a simple example of such a generalized space, which we name σ -topological spaces, or simply σ -spaces. Most of the basic concepts from topology carry over quite nicely to these new spaces, as we shall see.

1.1. The σ -Topology. Given a set X and a topology on X , $\mathcal{T}$, we call the pair $(X, \mathcal{T})$ a **topological space**. The sets $U \in \mathcal{T}$ are called the **open sets** of the space. The subtlety in the definition of a topology is that the collection $\mathcal{T}$ need be closed under *any* union of elements, even uncountable. This is of course motivated by the open ball topology, in particular the usual topology embedded in $\mathbb{R}$. It is no secret, however, that $\mathbb{R}$ and even any metric space (X, d) carry “nice” properties that may not survive generalizations. It is for this reason that we wish to slightly weaken the condition for unions: to allow more spaces.

Definition 1 (σ -Topology). *Let X be a set and $\mathcal{T}_\sigma$ a collection of subsets of X . The set $\mathcal{T}_\sigma$ is called a **σ -topology** on X if and only if*

- (i) $\emptyset, X \in \mathcal{T}_\sigma$.
- (ii) If $\{U_n\}_{n \in \mathbb{N}}$ is a countable collection of sets in $\mathcal{T}_\sigma$, then

$$U := \bigcup_{n \in \mathbb{N}} U_n \in \mathcal{T}_\sigma. \tag{1}$$

- (iii) If $U, V \in \mathcal{T}_\sigma$ then $U \cap V \in \mathcal{T}_\sigma$.

Now, given a set X and a σ -topology $\mathcal{T}_\sigma$ on X we call the pair $(X, \mathcal{T}_\sigma)$ a **σ -topological space**. It is apparent from this very definition that a topological space $(X, \mathcal{T})$ is necessarily a σ -topological space. Indeed, the only condition that was weakened was the closure under unions. If a topological space need be closed under possibly uncountable unions, it is certainly closed under countable unions. We shall often write $\mathcal{T}_\sigma$ to clarify that $(X, \mathcal{T}_\sigma)$ is a σ -topological space and not necessarily a topological space. The sets in $\mathcal{T}_\sigma$ are called the open sets of the space.

We shall now quickly introduce a σ -topological space that is not a topological space. Consider now $\mathbb{R}$ and the Lebesgue measure λ on $\mathbb{R}$. Denote by $\mathcal{M}$ the collection of Lebesgue measurable subsets of $\mathbb{R}$. Recall that $\mathcal{M}$ is a σ -algebra, i.e. it is closed under countable union, contains $\mathbb{R}$ and is closed under complements. Therefore, $\emptyset \in \mathcal{M}$ and (i) of Definition 1 is trivially satisfied. To see that (iii) also holds, note that given $U_1, U_2 \in \mathcal{M}$ the intersection $U_1 \cap U_2$ may be written as $U_1 \cup U_2^c$ where $U_2^c \in \mathcal{M}$ by closure under complement. The property then follows from closure under countable union.

Theorem 1.1. *There exists a σ -topological space $(X, \mathcal{T}_\sigma)$ that is not a topological space.*

Proof. Let $X = \mathbb{R}$ and $\mathcal{T}_\sigma = \mathcal{M}$ be the Lebesgue measurable sets in $\mathbb{R}$. Consider any Vitali set $V \subset \mathbb{R}$; then $V \notin \mathcal{M}$. We have already shown that $(X, \mathcal{M})$ is a σ -topological space. Suppose it is also a topological space. Then, any singleton $\{x\}$ is measurable, therefore open. By assumption, $\mathcal{M}$ is closed under union. Thus,

$$V = \bigcup_{x \in V} \{x\} \in \mathcal{M}$$

which we know to be impossible. Contradiction. $\square$

We shall return to this example later on, and study its behaviour. In fact, a major goal of this text is to study the σ -spaces induced by measurable spaces $(X, \mathcal{M})$.

1.2. Closed and Open Sets in σ -spaces. Here is where much of the usual topological notions carry over easily. Let $A \subseteq X$ be a set, where $(X, \mathcal{T}_\sigma)$ is a σ -topological space. A is said to be closed in $(X, \mathcal{T}_\sigma)$ if there exists an open set U such that $A = U^c$. We shall define further the **finish** of a set A .

$$\hat{A} = \bigcap_{F \supseteq A} F, \quad F \text{ is closed in } (X, \mathcal{T}_\sigma) \quad (2)$$

In a similar vein, we define the **visceral** of A , denoted $A_\circ$, to be:

$$A_\circ := \bigcup_{\mathcal{O} \subseteq A} \mathcal{O}, \quad \mathcal{O} \in \mathcal{T}_\sigma \quad (3)$$

The reader should note that these correspond to the closure and interior of a set $A \subseteq X$, in topology. We shall refer to them by different names in our setting since the visceral of a given set may not be open (and similarly the finish of such a set need not be closed). This is because an uncountable union of open sets may not be open; we shall see such an example later on. However, before we explore an example of such a space, we wish to study general σ -topological spaces.

Proposition 1.2. *Let $(X, \mathcal{T}_\sigma)$ be a σ -topological space and $\{F_n\}_{n \in \mathbb{N}}$ a countable family of closed sets in $(X, \mathcal{T}_\sigma)$. Then,*

- (i) $F := \bigcap_{n \in \mathbb{N}} F_n$ is closed in $(X, \mathcal{T}_\sigma)$.
- (ii) $F(k) = \bigcup_{n=1}^k F_n$ is closed for all $k \in \mathbb{N}$.

Proof. We first prove (i). To see this, we need only observe that $F^c = (\bigcap_{n \in \mathbb{N}} F_n)^c = \bigcup_{n \in \mathbb{N}} F_n^c$ which is open since each F_n^c is open and $\mathcal{T}_\sigma$ is a σ -topology (closed under countable unions).

For (ii), we write $F(k)^c = \bigcap_{n \leq k} F_n^c$ which is again open, as we have shown previously that $\mathcal{T}_\sigma$ is closed under finitely many intersections. $\square$

The following are direct from the usual notions of topological spaces. A point $x \in A \subseteq X$ is called an accumulation point, or limit point, of A if every neighbourhood $U \ni x$ intersects the set A at more points than simply $\{x\}$.

We now establish the notion of a subspace. Let $(X, \mathcal{T}_\sigma)$ be a σ -topological space and $Y \subseteq X$. The subspace σ -topology $\mathcal{T}_\sigma^Y$ induced by $\mathcal{T}_\sigma$ is the collection

$$\mathcal{T}_\sigma^Y := \{U \cap Y : U \in \mathcal{T}_\sigma\} \quad (4)$$

When referring to the subspace Y we really mean the pair $(Y, \mathcal{T}_\sigma^Y)$. The following lemma is now a useful, yet simple fact that carries over from topology:

Lemma 1.3. *Let $Y \subseteq X$. A is closed in $(Y, \mathcal{T}_\sigma^Y)$ if and only if there exists a closed set F in X such that $A = Y \cap F$.*

Proof. Suppose there exists a closed set F in X such that $A = Y \cap F$. Then, $X \setminus F$ is open with respect to $\mathcal{T}_\sigma$, and hence $Y \cap (X \setminus F)$ is open by definition of the subspace. By De Morgan's laws it follows that $Y \cap (X \setminus F) = Y \cap F^c$, where the complement is taken in X . However, we see $Y \setminus A = Y \cap F^c$ and hence $Y \setminus A$ is open so that A is closed in Y .

Conversely, assume A be closed in Y . Then $Y \setminus A$ is open in Y . There is then an open set $U \in \mathcal{T}_\sigma$ such that $Y \setminus A = Y \cap U$, and hence U^c is closed in X . We now claim that $A = Y \cap U^c$. Indeed, if $x \in A$ then $x \notin Y \setminus A$. In this case, $x \notin Y \cap U$ implying that $x \notin U$ and thus $x \in U^c$, or $x \in Y \cap U^c$. Moreover, if $x \in Y \cap U^c$ then $x \in Y$ and $x \notin U$ whereby we see $x \notin Y \cap U$. Thus, $x \notin Y \setminus A$. But, as $x \in Y$ we require $x \in A$ as well. This proves the lemma. $\square$

Theorem 1.4. *Let Y be a subspace of $(X, \mathcal{T}_\sigma)$ and let $A \subseteq Y$. Suppose that A and Y are closed relative to $\mathcal{T}_\sigma^Y$ and $\mathcal{T}_\sigma$, respectively. Then A is closed in $\mathcal{T}_\sigma$.*

Proof. By this lemma, we know that $A = Y \cap F$, for some closed set F in $(X, \mathcal{T}_\sigma)$. Now consider $X \setminus A = X \setminus (Y \cap F)$. Note that

$$\begin{aligned} X \setminus A &= X \setminus (Y \cap F) = X \cap (Y \cap F)^c \\ &= X \cap (Y^c \cup F^c) \\ &= Y^c \cup F^c \in \mathcal{T}_\sigma \end{aligned}$$

since Y^c and F^c are open. $\square$

Example 1.1. *Let $\mathcal{T}_\sigma := \{U \subseteq \mathbb{R} : U \text{ is countable or } U = \mathbb{R}\}$ and consider $(\mathbb{R}, \mathcal{T}_\sigma)$.*

We first claim that this is indeed a σ -topological space. Note that $\emptyset, \mathbb{R} \in \mathcal{T}_\sigma$ by construction. The intersection of countable sets is countable, and so (iii) holds. Similarly, the countable union of countable sets is again countable, so $\mathcal{T}_\sigma$ is indeed a σ -topology. This is certainly not a topological space and we now consider the set $A = [0, 1]$, neither open nor closed.

We claim that the visceral $A_\circ = A$. Certainly, by definition we see $\bigcup_{O \subseteq A} O \subseteq A$, where each $O \in \mathcal{T}_\sigma$ and therefore $A_\circ \subseteq A$. To see the converse, note that each $\{x\} \in \mathcal{T}_\sigma$ and hence $\{x\}$ is open for all $x \in A$. Thus, $x \in O$ for some O suitable for the union and

$$A_\circ = \bigcup_{O \subseteq A} O = A, \quad O\text{'s open}$$

But, A is not open! This proves that in σ -spaces the visceral does not always behave like the interior of a set, in a normal topological setting.

Consider furthermore the set A^c , again neither open nor closed in $\mathcal{T}_\sigma$. We seek the finish, $\widehat{A^c}$. Observe that

$$A^c = \bigcap_{O \subseteq A} O^c = \bigcap_{F \supseteq A^c} F, \quad F = O^c$$

and hence $\widehat{A^c} = A^c$ which is not closed!

We must impose more conditions to study the behaviour of sets in these spaces in a reasonable manner. However, the following are still obvious:

Proposition 1.5. *Let $A \subseteq X$ for a σ -space $(X, \mathcal{T}_\sigma)$. Then,*

- (i) *If A is open then $A_\circ = A$.*
- (ii) *If A is closed then $\widehat{A} = A$.*

1.3. Hausdorff σ -Spaces, Compactness and Continuous Functions.

We now impose an axiom on the smoothness of the topology $\mathcal{T}_\sigma$. As with regular topological spaces, we shall study, in this section, *Hausdorff* spaces. These are defined by:

Definition 2. A σ -topological space is said to be **Hausdorff** (or T_1) if for any two distinct points $x, y \in X$ there are sets $U_x \ni x$ and $U_y \ni y$, each in $\mathcal{T}_\sigma$, such that $U_x \cap U_y = \emptyset$.

It makes sense here to introduce the idea of a sequence. A collection $(x_n) \subseteq X$ is said to be a sequence in X . Now, we say $(x_n) \rightarrow x \in X$ as $n \rightarrow \infty$ if for any open set $U \ni x$ there exists N such that $x_n \in U$ for all $n \geq N$. In this case, we shall write

$$x = \lim x_n = \lim_{n \rightarrow \infty} x_n \text{ or } x_n \xrightarrow{n \rightarrow \infty} x$$

It is well known that the limit need not be unique in general topological spaces and hence cannot be the case for general σ -topological spaces either. However, it is true that Hausdorff spaces guarantee uniqueness of the limit, when it exists. We shall take this a step further. Recall that a directed set Λ is a collection equipped with a preorder such that every finite subset of Λ has an upper-bound.

Definition 3. Assume $(X, \mathcal{T}_\sigma)$ is a σ -topological space. Let Λ be a directed set and $x_\ell : \Lambda \rightarrow X$ a mapping. The collection $(x_\ell)_{\ell \in \Lambda}$ is called a **net**. We say the net (x_ℓ) converges to $x \in X$ if and only if for any neighbourhood $U \ni x$ there exists $\lambda \in \Lambda$ such that $x_\ell \in U$ for all $\ell \geq \lambda$.

We now have the following theorem:

Theorem 1.6 (Uniqueness of Limit of Nets). *Let $(x_\ell)_{\ell \in \Lambda}$ be a net on a σ -topological space $(X, \mathcal{T}_\sigma)$ and assume further that this space is Hausdorff. If $(x_\ell) \rightarrow x \in X$, then x is unique.*

Proof. By way of contradiction suppose there are two distinct $x, y \in X$ such that $(x_\ell) \rightarrow x$ and $(y_\ell) \rightarrow y$. Since our space is Hausdorff, we may take two neighbourhoods, U, V containing x, y respectively such that $U \cap V = \emptyset$.

By convergence in the net, there is λ_1, λ_2 such that $x_\ell \in U$ for all $\ell \geq \lambda_1$ and $x_\ell \in V$ whenever $\ell \geq \lambda_2$. Define $\delta \in \Lambda$ such that $\delta \geq \lambda_1$ and $\delta \geq \lambda_2$. Now, if $\ell \geq \delta$ one sees quite easily that $x_\ell \in U$ and $x_\ell \in V$, i.e. $U \cap V \neq \emptyset$, a contradiction. $\square$

The following nice result survives the move from topological spaces to σ -topological spaces:

Proposition 1.7. *Let $(X, \mathcal{T}_\sigma)$ be a σ -topological space and suppose $F \subseteq X$ is closed. If $(x_\ell)_{\ell \in \Lambda}$ is a net on F ($x_\ell \in F$ for all $\ell \in \Lambda$) and $(x_\ell) \rightarrow x \in X$ then $x \in F$.*

Proof. It is clear that F^c is open in $\mathcal{T}_\sigma$. Suppose that $x \notin F$. For ℓ “sufficiently large” we have $x_\ell \in F^c$ by convergence. But then $F \cap F^c \neq \emptyset$. $\square$

We now turn briefly towards a discussion of compactness.

Definition 4. Let $(X, \mathcal{T}_\sigma)$ be a σ -topological space. We shall call this space **compact** if every open covering of X has a finite sub-covering. That is, if $\{U_\alpha : \alpha \in I\}$ is an indexed collection of sets in $\mathcal{T}_\sigma$ such that

$$X \subseteq \bigcup_{\alpha \in I} U_\alpha$$

then there exists a **finite** set $J \subseteq I$ such that

$$X \subseteq \bigcup_{\beta \in J} U_\beta$$

It is a well known result that if $(X, \mathcal{T})$ is a Hausdorff topological space and $K \subseteq X$ a compact subspace, it is closed in X . This is simply not true for the σ -topological Hausdorff spaces in general. We show this below.

First we give a definition for subspaces. If $(X, \mathcal{T}_\sigma)$ is a σ -topological space and $Y \subseteq X$ a subspace we call Y compact in X if every covering of Y by elements in $\mathcal{T}_\sigma$ has a finite sub-covering. We are now equipped to disprove something that held true for general Hausdorff topological spaces.

Example 1.2. Let $\mathcal{Z}$ denote the set of all null sets in $\mathbb{R}$ together with $\{\mathbb{R}\}$. There exists a compact subspace $K \subset \mathbb{R}$ such that K is not closed in X .

We solve this problem as follows. First, it is trivial to check that $(\mathbb{R}, \mathcal{Z})$ is a σ -topological space and it is left to the reader to convince themselves of this fact. Of course, every singleton $\{x\}$ is open where $x \in \mathbb{R}$ and thus it is clear that this space is also Hausdorff. Consider now the set $\{0\} \subset \mathbb{R}$. This is certainly compact, as it is finite and hence any open covering by open sets in $\mathcal{T}_\sigma$ must have a finite sub-covering.

Now, $\{0\}$ is open and not closed. If it were closed, then $\mathbb{R}^\times$ (the units of $\mathbb{R}$) would be a null set, which is absurd since it contains an open interval with respect to the Euclidean topology.

By now it should be clear that σ -topological spaces behave much more erratically than their topological counterparts. This comes from the restriction to countable unions, and the difference is astounding.

We now turn towards continuous functions. From hereon we shall denote by $(X, \mathcal{T}_\sigma)$ and $(Y, \mathcal{W}_\sigma)$ two σ -topological spaces. A mapping $f : X \rightarrow Y$ is called **continuous** if for every set $V \in \mathcal{W}_\sigma$ the pre-image $f^{-1}(V)$ is open in $(X, \mathcal{T}_\sigma)$.

It still holds that continuous functions preserve compactness.

Theorem 1.8. Let $f : X \rightarrow Y$ be continuous and $K \subseteq X$ compact. Then, $f(K)$ is compact in $(Y, \mathcal{W}_\sigma)$.

Proof. Let $\{\mathcal{H}_\alpha\}_\alpha$ be an open covering of $f(K)$. That is, we assume $f(K) \subseteq \bigcup_\alpha \mathcal{H}_\alpha$. Now, this implies that $K \subseteq f^{-1}(\bigcup_\alpha \mathcal{H}_\alpha) = \bigcup_\alpha f^{-1}(\mathcal{H}_\alpha)$. By compactness, we find that $K \subseteq \bigcup_{\alpha \in A} f^{-1}(\mathcal{H}_\alpha)$ for some finite index set A .

Hence,

$$f(K) \subseteq f\left(\bigcup_{\alpha \in A} f^{-1}(\mathcal{H}_\alpha)\right) = \bigcup_{\alpha \in A} \mathcal{H}_\alpha$$

This completes the proof. $\square$

Suppose now that $(X, \mathcal{T}_\sigma)$ is a compact space and $Y \subseteq X$ is a closed subspace. We claim that Y is compact in X . To see this, let $\{\mathcal{G}_\alpha\}_\alpha$ be an open covering of Y in $\mathcal{T}_\sigma$: $Y \subseteq \bigcup_\alpha \mathcal{G}_\alpha$ where $\mathcal{G}_\alpha \in \mathcal{T}_\sigma$ for all α . Now, $X \setminus Y$ is open and so,

$$\bigcup_\alpha \mathcal{G}_\alpha \cup Y^c \supseteq X, \quad Y^c = X \setminus Y$$

defines an open covering of X by elements of $\mathcal{T}_\sigma$. Since X is a compact space, it follows that there is a finite index set A such that

$$X \subseteq \bigcup_{\alpha \in A} \mathcal{G}_\alpha \cup Y^c$$

In particular, we find $Y \subseteq \bigcup_{\alpha \in A} \mathcal{G}_\alpha$. Since $Y \cap Y^c = \emptyset$ one deduces easily that $Y \subseteq \bigcup_{\alpha \in A} \mathcal{G}_\alpha$ and hence that Y is compact in $(X, \mathcal{T}_\sigma)$.

2. THE LEBESGUE σ -TOPOLOGICAL SPACE

In this chapter we shall consider the σ -topological space induced by the Lebesgue measure on $\mathbb{R}^d$, $d \geq 1$. Let $\mathcal{M}$ denote the collection of all Lebesgue measurable sets on $\mathbb{R}^d$. This is a σ -algebra by construction. We have already shown (it is quite easy to do so) that $\mathcal{M}$ is also a σ -topology on $\mathbb{R}^d$. The space $(\mathbb{R}^d, \mathcal{M})$ is what we shall call the **Lebesgue σ -topological space**.

OBSERVATIONS:

- A subset of $\mathbb{R}^d$ is either clopen or neither open nor closed. Certainly, suppose that $A \subseteq \mathbb{R}^d$ is not clopen. Then, either it is closed but not open, open and not closed or neither closed nor open. However, since $\mathcal{M}$ is closed under complements the first options cannot ever occur and hence we find that A is neither open nor closed.
- $(\mathbb{R}^d, \mathcal{M})$ is Hausdorff since every singleton is open.
- $(\mathbb{R}^d, \mathcal{M})$ is not a topological space.

Theorem 2.1 (Characterization of Compact Sets). *Let $K \subseteq \mathbb{R}^d$. The following are equivalent:*

- (i) K is compact in $(\mathbb{R}^d, \mathcal{M})$.
- (ii) K is finite.

Proof. (i $\implies$ ii). We shall show that if K is infinite then it cannot be compact. Indeed, every singleton $\{x\}$ is open relative to $\mathcal{M}$ and hence given an infinite set K the collection $\{x\}_{x \in K}$ is an open covering of K for which no finite sub-covering can exist.

(ii $\implies$ i). If K is finite, and $\{U_\alpha\}_\alpha$ an open covering of K we enumerate $K := \{x_1, \dots, x_n\}$. We see then that $\{x_1, x_2, \dots, x_n\} \subseteq \bigcup_\alpha U_\alpha$. Pick α_j so that $x_j \in U_{\alpha_j}$ for $j = 1, 2, \dots, n$ and $\{U_{\alpha_j}\}_{j=1}^n$ is a finite sub-covering. $\square$

Theorem 2.2. *Let $K \subseteq \mathbb{R}^d$. The following are equivalent:*

- (i) K is compact in $(\mathbb{R}^d, \mathcal{M})$.
- (ii) Every sequence in K has a convergent subsequence.

Proof. (i $\implies$ ii). By the previous theorem K is finite. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in K . Then, it must contain a constant, and hence convergent, subsequence.

(ii $\implies$ i). Suppose now that (i) is false: i.e. assume K is infinite. In this case, we may extract a countable subset $\{x_n : n \in \mathbb{N}\}$ of K . To see that this can be done, for $n \in \mathbb{N}$ let S_n consist of n distinct elements in K . Then take the set $\bigcup_{n=1}^\infty S_n$. Now, this collection of x_n consists of distinct elements and hence defines a sequence of infinitely many distinct points. We show that no subsequence can converge to any $x \in K$.

Certainly, suppose that $(x_{n_k}) \rightarrow x \in K$ as $n_k \rightarrow \infty$. The singleton $\{x\}$ is open, and hence is a neighbourhood of x . By convergence, there exists an index N so large that $x_{n_k} \in \{x\}$ for all $n_k \geq N$. This contradicts the construction of the sequence: as all elements were taken to be distinct. $\square$

An interesting proposition follows, which completely characterizes convergence of nets:

Proposition 2.3 (Characterization of Nets). *Let $(x_\ell)_{\ell \in \Lambda}$ be a net on $(\mathbb{R}^d, \mathcal{M})$. Then $(x_\ell)_{\ell \in \Lambda}$ is convergent if and only if it is eventually constant.*

Proof. One direction is trivial. Suppose that $(x_\ell)_{\ell \in \Lambda}$ is convergent, with some limit point $x \in \mathbb{R}^d$ (which is unique by our observation that this space is Hausdorff). Since $\{x\}$ is open in our σ -topological space, there is $\lambda \in \Lambda$ such that for all $\ell \geq \lambda$ one has $x_\ell \in \{x\}$, i.e. $x = x_\ell$. Hence, this net is eventually constant. $\square$

We now have the following characterization theorem for the *entire* space.

Theorem 2.4. *Let $(X, \mathcal{T}_\sigma)$ be a σ -topological space. The following statements are equivalent:*

- (i) The visceral of every subset of X is open in $\mathcal{T}_\sigma$.
- (ii) $(X, \mathcal{T}_\sigma)$ is a topological space.

Proof. (i $\implies$ ii). Suppose not, then since $\mathcal{T}_\sigma$ is a σ -topology there exists an uncountable collection $\{G_\alpha\}_\alpha$ of sets in $\mathcal{T}_\sigma$ such that $\bigcup_\alpha G_\alpha \notin \mathcal{T}_\sigma$. Now, if we let H denote this union, we have by hypothesis that $H_\circ$ is open in $\mathcal{T}_\sigma$. That is, for a set $\mathcal{O} \subseteq \bigcup_\alpha G_\alpha$ one has

$$\bigcup_{\mathcal{O} \subseteq H} \mathcal{O} \in \mathcal{T}_\sigma$$

We note that we immediately have the inclusion

$$\bigcup_{\mathcal{O} \subseteq H} \mathcal{O} \subseteq \bigcup_{\alpha} G_{\alpha}$$

Let now $x \in \bigcup_{\alpha} G_{\alpha}$. Then, for a particular γ one sees $x \in G_{\gamma}$, which lives in $\mathcal{T}_{\sigma}$. Note that $G_{\gamma} \subseteq H$ by definition of H , so that $\mathcal{O} = G_{\gamma}$ for some open set in the visceral. That is, $x \in G_{\gamma} \subseteq \bigcup_{\mathcal{O} \subseteq H} \mathcal{O}$. We then conclude that $\bigcup_{\mathcal{O} \subseteq H} \mathcal{O} = \bigcup_{\alpha} G_{\alpha}$. This is a contradiction.

(ii $\implies$ i). In a topological space the visceral corresponds to the interior of the set, which is always open. □

One may alternatively use the following characterization:

Corollary 2.5. *Let $(X, \mathcal{T}_{\sigma})$ be a σ -topological space. The following statements are equivalent:*

- (i) *The finish of every subset of X is closed in $\mathcal{T}_{\sigma}$.*
- (ii) *$(X, \mathcal{T}_{\sigma})$ is a topological space.*

Proof. It is sufficient to show that (i) holds if and only if (i) of Theorem 2.4 holds. Indeed, suppose there exists a set $A \subseteq X$ with non open visceral $A_{\circ}$ with respect to the σ -topology. That is, $\bigcup_{\mathcal{O} \supseteq A} \mathcal{O} \notin \mathcal{T}_{\sigma}$ where the $\mathcal{O}$'s are open. If we let F denote a closed set containing A^c , then by assumption we know the finish $\bigcap_F F$ is closed and therefore $\bigcup_F F^c$ is open. Since this union is taken over the $F \supseteq A^c$ closed, we have $F^c \subseteq A$ where F^c is open. In fact, it follows that

$$\widehat{A^c}^c = \bigcup_{F \supseteq A^c} F^c = \bigcup_{F^c \subseteq A} F^c = A_{\circ} \quad (5)$$

Since the finish of A^c is closed, the complement $\widehat{A^c}^c \in \mathcal{T}_{\sigma}$ whence we find $A_{\circ} \in \mathcal{T}_{\sigma}$ by equality: a contradiction. A symmetric argument proves the converse. □

References

- (1) Stein & Shakarchi. Real Analysis: Measure Theory, Integration and Hilbert Spaces. Princeton Lectures in Analysis, Princeton Press 2003.
- (2) Topology, James Munkres.